

# TRINETRA

Evidence-first, physics-grounded AI detection & classification of exoplanet transits in noisy TESS light curves

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Bharatiya Antariksh Hackathon 2026 — Problem Statement 7

Team Name: [TEAM NAME]

Team Leader: [LEADER NAME]

*AI-enabled Detection of Exoplanets from Noisy Astronomical Light Curves*

# Team

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| Role        | Name   | College / Institution |
|-------------|--------|-----------------------|
| Team Leader | [Name] | [College]             |
| Member 1    | [Name] | [College]             |
| Member 2    | [Name] | [College]             |
| Member 3    | [Name] | [College]             |

*Suggested roles: data/conditioning · ML/classifier · physics/parameter-fit · visualization/report*

# Opportunity

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## **Different from existing ideas**

Most pipelines detect OR classify, often with uncalibrated scores. TRINETRA is evidence-first + physics-grounded, fusing interpretable physics features with a CNN, on a spine already benchmarked vs full TLS.



## **How it solves the problem**

Detect periodic dips → classify (transit/eclipse/blend/other) with explicit crowded-field/blend discriminators → attach SNR + calibrated confidence → fit period/depth/duration → visualize.



## **USP**

Hybrid physics+DL (interpretable, small-data robust); calibrated confidence by construction; physics decides detection (recall-prioritized); explicit blend handling; extends a validated, pre-registered spine.



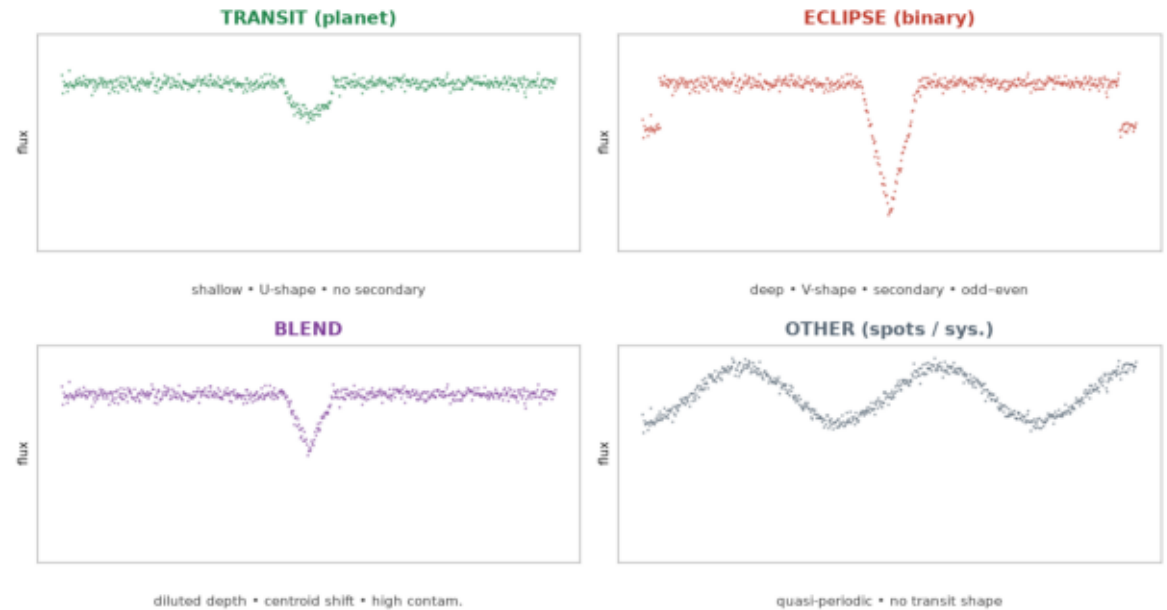
## **Already demonstrated on fresh MAST data**

Pi Mensae c blindly recovered (P=6.262 d, 289 ppm, SDE 12.3); TIC 100029948 flagged as an eclipsing binary (depth ~25%, odd-even 0.23).

# Features

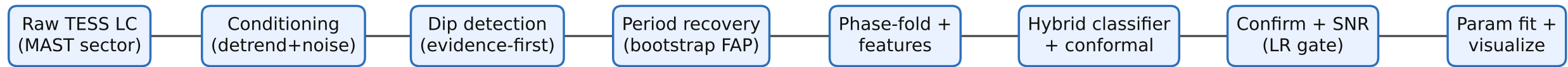
- Robust periodic-dip detection in noisy
- crowded fields
- 4-class AI classification: transit / eclipse / blend / other
- 5 $\sigma$  SNR significance per event
- Transit parameter fit: period, depth, duration (+ uncertainties)
- Calibrated confidence + annotated
- visualizations
- Reproducible; benchmarked vs full TLS

Four signal classes — distinct physical signatures the classifier separates



# Process Flow

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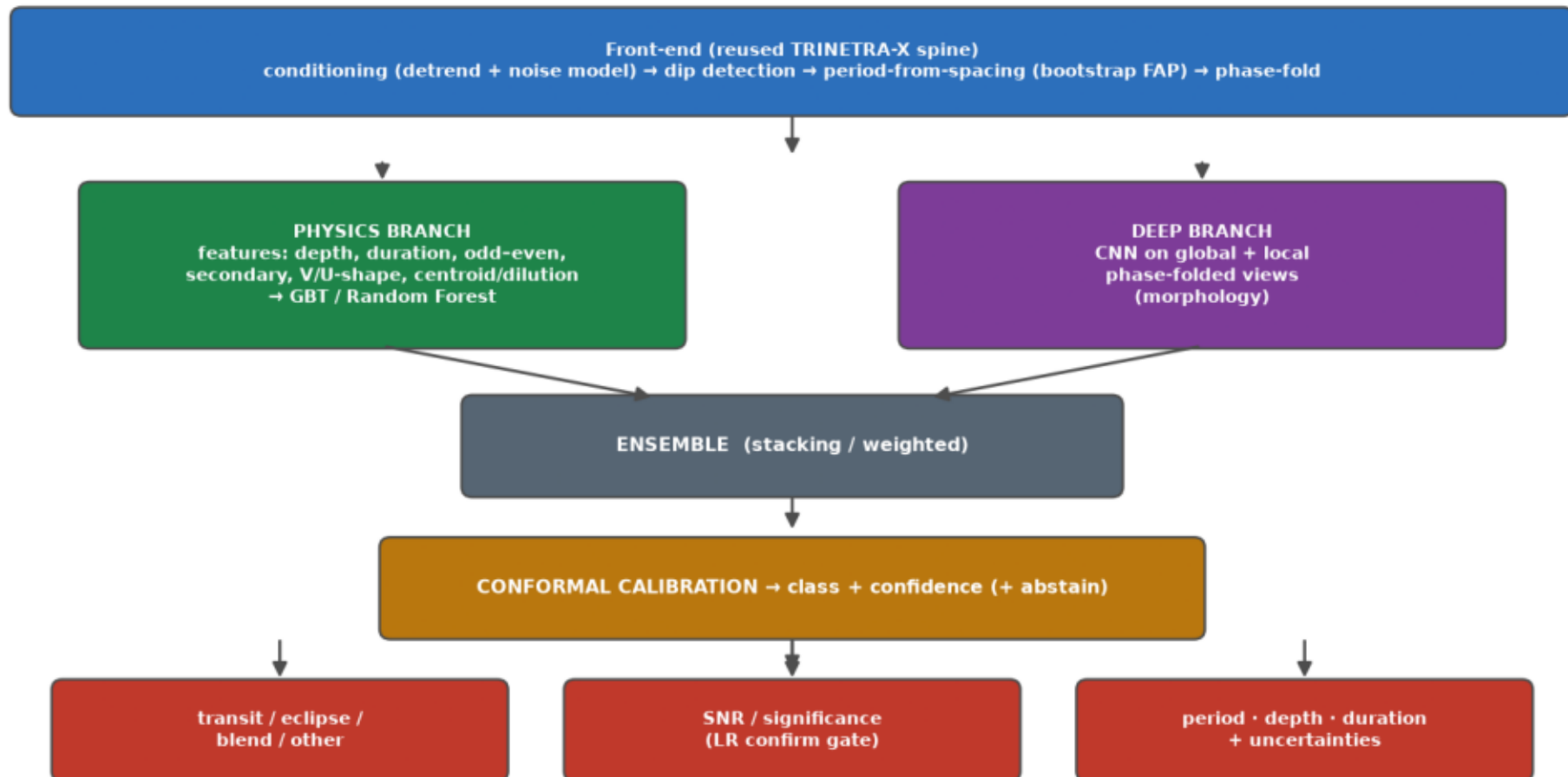


Front-end (conditioning → detection → period recovery) reused from the validated TRINETRA-X pipeline;  
classifier + blend features + visualization are the new build.

# Architecture

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## TRINETRA — Hybrid Detection & Classification Pipeline



# Technologies

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- **Core scientific Python**  
numpy · scipy · pandas · matplotlib
- **Astronomy**  
lightkurve · astropy · astroquery/MAST · wotan · transitleastsquares · batman
- **Machine learning**  
scikit-learn · XGBoost/LightGBM (physics branch) · PyTorch (CNN branch) · conformal (MAPIE)
- **Method**  
injection-recovery into real LCs · bootstrap FAP · conformal prediction · benchmark vs full TLS
- **All open-source**  
no specialized or licensed software; data is public (MAST)

# Estimated Implementation Cost

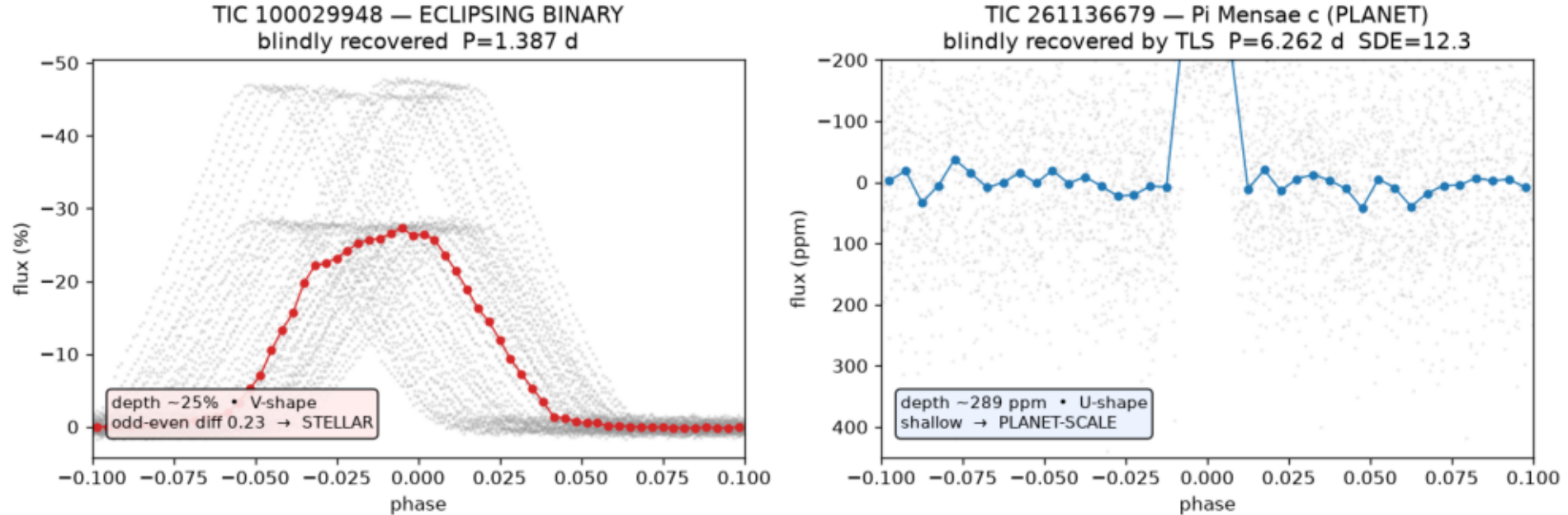
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- **Software: ₹0**  
entirely open-source stack
- **Data: ₹0**  
public TESS archive (MAST); one sector ~40–55 GB raw, kept slim by condition-then-discard
- **Compute**  
standard laptops for dev + a GPU session for CNN training; optional free cloud credits
- **Net**  
negligible cost — feasibility is high; the spine already runs on real data



# Proof of Concept — fresh MAST data (validated 2026-06-26)

Transit vs Eclipse — fresh MAST data → conditioning → recovery → physics discrimination (PS7)



Left: TIC 100029948 — eclipsing binary flagged by physics features (depth ~25%, V-shape, odd-even 0.23).

Right: Pi Mensae c — blindly recovered (P=6.262 d vs lit. 6.268; depth 289 ppm vs lit. ~315; SDE 12.3).

**The detection spine works end-to-end today; only the trained 4-class classifier remains to be built.**